ÁLGEBRA

— YisusReveck —

$$a^0 = 1 \qquad a^{-n} = \frac{1}{a^n}$$

$$\frac{a^m}{a^n} = a^{m-n} \qquad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$a^m \cdot a^n = a^{m+n} \qquad \sqrt[m]{a}\,\sqrt[n]{a} = a^{\frac{1}{m}+\frac{1}{n}}$$

$$(ab)^n = a^n b^n \qquad \sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \qquad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$(a^m)^n = a^{mn} \qquad \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

$$\log_b(xy) = \log_b x + \log_b y$$

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

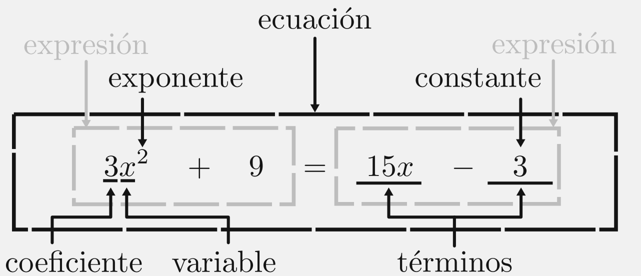
$$\log_b(x^k) = k\log_b x$$

$$\log_b x = \frac{\log_k x}{\log_k b}$$

$$\ln x = \log_e x$$

$$\ln(e^x) = x \qquad e^{\ln x} = x$$

$$\log_b 1 = 0 \qquad \log_b b = 1$$



π	$\approx$	3.14159
e	$\approx$	2.71828
$\sqrt{2}$	$\approx$	1.41421
φ	$\approx$	1.61803
τ	$\approx$	6.28318

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta = b^2 - 4ac$$

$\Delta > 0$	2 soluciones reales
$\Delta = 0$	1 solución real
$\Delta < 0$	0 soluciones reales

$$(a+b)(a-b) = a^2 - b^2 \qquad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$(a+b)^2 = a^2 + 2ab + b^2 \qquad (a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

$$(a-b)^2 = a^2 - 2ab + b^2 \qquad (x+a)(x+b) = x^2 + (a+b)x + ab$$